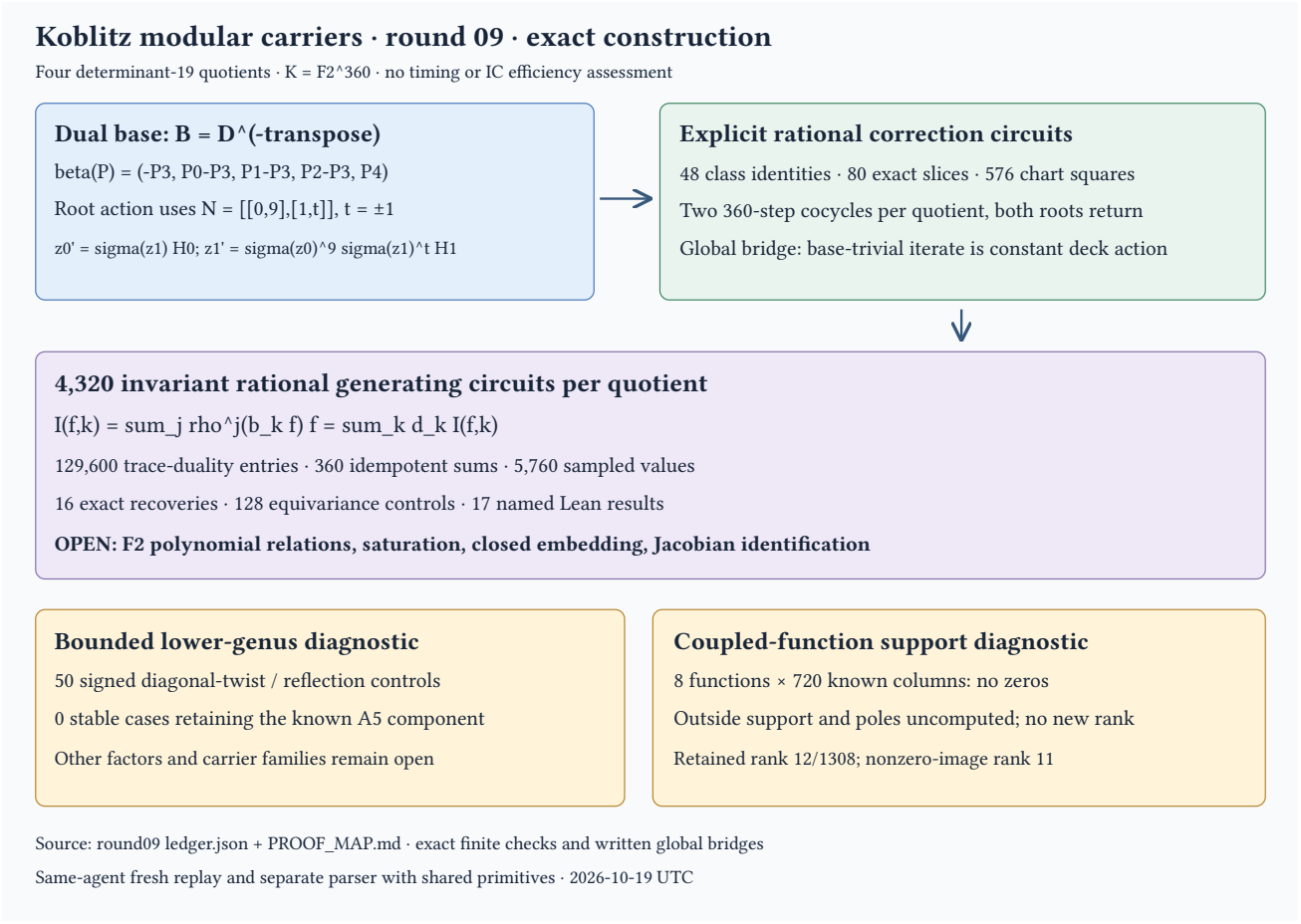
Round09 — explicit Frobenius and invariant rational generators

2026-10-19 UTC. Parent92688745f0ffadf854367ad1d2f33fa05bf7ce2a. The four determinant19 quotients now have explicit Frobenius coordinate circuits and4,320 invariant rational generating circuits each. This is a function-field construction. F2 polynomial relations, denominator saturation and a closed projective embedding remain open. Relation ranks are unchanged; no timing or IC efficiency conclusion is made.



Quotient descent: what is explicit

Use K=F2^360 with the frozen modulus z^360+z^5+z^3+z^2+1 and the four round08 graph kernels. Each split chart has five elliptic factors and two Kummer coordinates, z_i^19=product_j f_Tij(P_j), with40 actual polynomial Miller functions across the four graphs. The six charts and200 transition factors remain frozen inputs.

The ordinary augmentation source uses D, but the dual chart base uses B=D^(-transpose). Its action is

```
beta(P)=(-P3,P0-P3,P1-P3,P2-P3,P4).
alpha_base(P)=beta(sigma(P)), sigma(a)=a^2 on K.
```

The product principal polarization used over K does not identify the two descent actions. The pairing identity and both inverse matrix products are proved in Lean. B^5=1 is checked there as well.

For graph generators in the order(Frob S,S), the actual point checks give M=[[2t,1],[-2,0]], with t=±1. Its signed inverse modulo19 is N=[[0,9],[1,t]]. The root lift is

```
z0'=sigma(z1)*H0(sigma(P)),
z1'=sigma(z0)^9*sigma(z1)^t*H1(sigma(P)).
```

For t=-1 the last factor is an inverse. All H factors are retained as explicit coefficient arrays and fixed-point line/group rational circuits; there is no uncomputed correction symbol. The exact formula, constants, fixtures, slices and chart transport are in q7-Frobenius-coordinate-circuits.json. The written derivation and its formalization limits are in PROOF_MAP.md.

Check	Total	Scope
Univariate class/sum identities	48	Exact rational-function equality
Coefficient-Frobenius Miller identities	40	All actual factors
One-variable full-lift slices	80	Two fixtures, five variables, two roots, four graphs
Generator semilinear steps	2,880	Two360-step cocycles per graph
Parsed semilinear steps	2,880	Recomposition with shared primitives

Check	Total	Scope
Chart squares	576	All36 chart pairs, both roots and fixtures
Old transition coefficient checks	400	All200 factors, both fixtures

Both roots and every base point return after360 steps. This alone is a pointwise check. The global argument first proves that the360-fold lift is a base-trivial automorphism of the connected Kummer function field, hence a constant mu19^2 deck action. A nonzero-root fixture then determines both constants to be1. The second fixture is an additional control. The global connectedness/Picard bridge is written, not fully Lean.

An explicit invariant generating family

Let b_k=z^k, k=0..359, and d_k be its trace-dual basis. Exact binary linear algebra computes d_k; native checks verify129,600 trace-pairing entries,360 separability-idempotent sums and64 direct trace controls. For each of the twelve chart coordinates f, retain the circuit

```
I_(f,k)=sum_(j=0..359) rho^j(b_k*f),
rho^j(f)(P)=sigma^j(f(alpha^(-j)(P))),
f=sum_(k=0..359) d_k*I_(f,k).
```

The rho orbit sum is fixed, and the idempotent identity gives recovery. This supplies4,320 possibly redundant rational generators per quotient, not4,320 algebraically independent coordinates. The constant extension of the field generated by these invariants recovers the original function field. Galois descent supplies the fixed-field interpretation; see the written proof and Stacks Galois descent. Trace duality uses the nondegenerate separable trace pairing described in Stacks trace and norm.

All360 weights are evaluated for x0,y0,z0,z1 at one fixture per graph: 5,760 invariant values,16 exact coordinate recoveries and128 equivariance controls pass, with1,440 additional point steps. At a general K-point, the correct identity is I(alpha(P))=I(P)^2. Its value need not belong toF2. The universal recovery formula applies to all twelve functions; only the four specified coordinate families are evaluated in this point control. No invariant polynomial relation ideal or saturation has been computed.

Bounded lower-genus quotient diagnostic

Test five diagonal twists g^u and five reflections g^j R, for both traces: 50 signed controls on the known augmentation component. Here g=D^2, R D R=D^-1 and D R D^-1=g R. Twisted Frobenius on that component is D^(2u+1) tensor the elliptic Frobenius companion.

A reflection is stable exactly when2u+1=0 mod5, hence u=2. That twist collapses the known component to E_t^4; its reflection-fixed part has polynomial(E_t)^2 and dimension2. The other twists retain the known four-dimensional A5 component but no tested reflection is stable. Thus zero of50 controls gives both properties. Exact integer characteristic polynomials and all commutators are checked. This does not exclude A5 in other Jacobian factors, or other lower-genus carrier families.

The carrier genus is2841. Conditional Riemann-Hurwitz arithmetic gives reflection-quotient genus≤1421; its actual genus, ramification and equations are not computed. The inequality is proved in Lean with its hypothesis.

Direct relations: eight coupled functions

M12 restricts x0+xinf+c, y0+yinf+c, x0*xinf+c and x0*yinf+xinf*y0+c, c=0,1, to all720 known norm-family columns (indices588..1307). These are actual target coordinates after the deck maps in split or degree11 coordinate fields. All40,320 component/function evaluations agree within each closed-column cycle. Every one of the eight functions has zero known columns in this population.

This records support information, not complete principal divisors. Zeros outside these columns, their multiplicities and poles remain uncomputed. No new relation or rank is promoted. The retained evaluated matrix has rank12 on1308 columns; the nonzero-image matrices have rank11 on1165/1160 columns. General Jacobian reduction and individual descent remain open.

Evidence, failures and next experiments

The ledger records each requirement as partial or not attempted at its actual scope. Five new JSON records replay byte for byte; the separate Q7 parser checks the recorded correction functions and chart action. Round08 replays unchanged. Seventeen new named Lean results pass the pinned kernel build and transitive axiom audit. Two source-bound universal identities each receive64 issuer and128 extra evaluation checks. Native finite-field semantics and the global geometric arguments remain outside those Lean statements. Same-agent checks share declared primitives; there is no external audit.

The archive preserves the initial Rust parsing error, the stale-binary output-path rejection and the failed Lean recovery statement missing a DecidableEq parameter. No accepted result uses them. A wrong relative append path left one intermediate build checking only the earlier source; that receipt is labeled separately. Final checks are in VALIDATION.md. Earlier archives remain unchanged. The main-branch merge only resolves blocked verifier metadata; its receipt is retained, and no executable verification command was disabled.

Next quotient experiment: reduce the trace family using a Frobenius-adapted base, then compute an invariant relation ideal on a specified open chart with exact inverse substitutions and denominator saturation. Verify gluing and the retained polarization on the descended model. Next relation experiment: compute complete outside zero support and poles for a coupled function before joining new columns and reporting rank. Next lower-genus experiment: compute actual reflection quotient geometry or test other Jacobian components/families. The bounded negatives above are not nonexistence theorems. The user makes the IC efficiency assessment; this round contains no timing or solver measurement.

Formal proof map and written arguments

Kernel-checked statements

KoblitzRound09.lean imports the pinned round08 project. Seventeen named new theorems are audited transitively. Only the standard propext, Classical.choice and Quot.sound axioms occur; no sorryAx, custom axiom or native trust. The proof statements have the following exact scopes.

Theorems	Statement scope
dual_pairing_identity	Universal commutative-ring pairing identity
dual_matrix_left, dual_matrix_right, dual_matrix_order	Explicit integer matrices, inverse products and fifth power
root_matrix_inverse	All t in ZMod19, both products equal1
rotation_reflection_commute, stable_twist	Finite ZMod5 permutation and twist conditions
weil_five_plus, weil_five_minus	Universal degree10 factorizations for both signs
pow_nine_nineteen, root_lift_plus, root_lift_minus	Universal power identity and conditional root equations
rooted_deck_phase	Nonzero rooted fixture forces a constant phase to1
chart_transport_cancel	Nonzero-denominator chart correction cancellation
reflection_genus_bound	Integer implication from the specified RH inequality
trace_orbit_invariant	Additive action fixing a sum by permuting its summands
trace_dual_recovery	Finite double-sum recovery under an explicit idempotent hypothesis

Two IDC certificates bind dual_pairing_identity and weil_five_plus to the exact Lean source hash, with64 issuer and128 extra checks each. They certify their universal polynomial identities, not the entire geometry.

Mixed Miller identity — written argument

Fix a nonzero19-torsion point T of one retained elliptic factor and choose the normalized Miller function f_T with divisor19(T)-19(O). Let g_A,B be the normalized line quotient with divisor(A)+(B)-(A+B)-(O), including the vertical/tangent cases by rational continuation. Put

```
h_T(X,Y)=g_(T-Y,Y)(X)/(x(X)+x(Y)).
```

The divisor in X is(T-Y)+(O)-(T)-(-Y), since the vertical denominator has divisor(Y)+(-Y)-2(O). The ratio f_T(X+Y)/(f_T(X)*f_T(Y)) has19 times this divisor. Their quotient is constant in X. The standard local parameter at O and the monic normalization of f_T and the line quotient make that constant1. This proves the generic rational identity

```
f_T(X+Y)=f_T(Y)*h_T(X,Y)^19*f_T(X).
```

It holds as a rational identity across charts. The native code stores the group/line circuit for h, checks both orientations in exact one-variable function fields, and checks80 full lift slices. Slices alone do not prove the global identity. The divisor/normalization argument is the bridge; it is not formalized in Lean.

Integer combinations of sigma(T) have explicit univariate corrections prod f_T^weights=f_sum*g^19. All48 class/sum identities are checked by cross-multiplying reduced rational functions on the elliptic curve. The actual graph equality D*sigma(T)=T*M, inverse matrix N and beta=B give

```
H_i(P)=sum_correction_i(P3)
  *product_(j=1..3) h_Tij(P_(j-1),-P3)
  /product_(j=0..4) class_correction_ij(Pj).
```

Consequently H_i(P)^19 is exactly the ratio between the destination radicand at beta(P) and the signed N-combination of coefficient-Frobenius radicands at P. This proves the displayed root lift generically. All coefficients of the univariate factors, T points and rational operations are retained; sigma and beta are explicit, so H is an executable circuit.

Global cocycle and chart transport — written argument

The retained independent19-torsion Picard classes give the connected, integral Kummer extension of degree19^2, as in round08's written identification with the actual quotient. Since N is invertible modulo19, the lift is a generic isomorphism over the base Frobenius. Its360-fold iterate fixes the base: B^5=1 and sigma^360=1. It is therefore an automorphism of that Kummer function field over its base. Because mu19 is contained in K, each z_i can only acquire a constant mu19 multiplier. A fixture with both roots nonzero, and with every iteration defined, fixes both multipliers to1. Two such fixtures per graph pass. Lean proves the nonzero-root cancellation implication, not this whole geometric bridge.

Let rho_i,k be the product of the five retained chart transition factors, with rho_i,0=1. In source chart k and target chart l, the lifted formula is

```
z'_i,l=product_j sigma(z_j,k)^N_ji * H_i(sigma(P))
  *rho_i,l(B*sigma(P))/product_j sigma(rho_j,k(P))^N_ji.
```

This is substitution and cancellation of the chart definitions. All200 old factors pass coefficient-Frobenius checks at both fixtures. The parsed checker verifies every36 chart pair, both roots and both fixtures, 576 squares total, and recomposes2,880 semilinear steps using shared field primitives. Full global identities use the written rational argument.

Invariant rational generators — written argument

The exact field basis b_k and its trace dual d_k satisfy sum_k d_k*sigma^j(b_k)=delta_(j,0). Native arithmetic verifies all360 identities directly, in addition to129,600 trace-duality entries and64 trace controls. Let L be the retained geometrically integral quotient function field over K and rho its semilinear function action of order360. The fixed trace sum I_(f,k)=sum_j rho^j(b_k*f) is invariant by cyclic reindexing. Expanding the double sum gives

```
sum_k d_k*I_(f,k)
  =sum_j (sum_k d_k*sigma^j(b_k))*rho^j(f)=f.
```

Lean proves the reindexing and this double-sum implication for arbitrary finite index types; the native field/basis interpretation remains separate. Taking f to be all twelve chart coordinates shows that adjoining K to the field E generated overF2 by these invariants gives L. To identify E with L^rho, use the linear independence of the K basis after scalar extension from the invariant field: finite Galois descent gives K tensor_F2 L^rho ≅ L as vector spaces. Expressing an invariant in L=K E uniquely in that basis forces all coefficients except the constant one to vanish. Hence L^rho=E. This is a written application of Galois field/vector-space descent, not a generated polynomial presentation.

The point controls compute5,760 invariant values for four coordinate families,16 recoveries and128 identities I(alpha(P))=I(P)^2. They are not a claim that arbitrary K-point invariant values belong toF2. The12×360 family is a rational generating family, potentially redundant. Its polynomial relation ideal, saturation, gluing of invariant affine opens and closed projective embedding are not computed.

Low-genus and relation limits

L7 checks all50 signed cases in one known eight-dimensional Tate-module representation. Integer traces/Newton identities give exact characteristic polynomials; the stable reflection-fixed part is E_t^2. The finite Lean lemmas support the twist and commutator calculations. No claim covers other Jacobian factors. The RH lemma assumes the inequality and does not compute ramification, the quotient genus or equations.

M12 evaluates eight coupled functions on the720 known norm columns only. Each coordinate-field calculation checks all deck-cycle components and zero consistency. No outside zero, multiplicity or pole calculation has been supplied. There is no new complete principal relation or rank. The earlier odd reverse divisor and independent forward norms remain frozen; general reduction and individual descent are still open.

Sources and trust boundary

The dual-isogeny and Picard framework is explained in Milne, Abelian Varieties. The Kummer framework is in Stacks Kummer theory. For the field/vector-space descent used above, see Stacks Galois descent; for separable trace duality, see Stacks trace and norm. These references support the frameworks; the explicit circuits and application to the four retained graphs are this study's written/native construction. The whole construction is not a Lean theorem. The native field semantics, connected-cover identification, global divisor geometry and modular-factor geometry remain explicit trust boundaries.